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Phase 01: Formulation lock-in and application scope

This document is Phase 01's deliverable: one precise statement of the governing equations, the DOF layout, and the two target test problems, per docs/ROADMAP.md's "Ready for 02" checklist. Everything below is built from Maxwell's equations (public, textbook material), the potential definitions already in use throughout this project (see docs/OPEN_BOUNDARY_ABC.md), and the two already-cited papers behind the all-frequency-stable decision (Zhao & Fu 2017, Yan 2021) – no employer-authored material enters this derivation, per the roadmap's formulation source boundary.

1. Governing equations (all-frequency form)

1.1 Starting point: Maxwell's equations and the potential definitions

Time-harmonic fields with the $e^{+j\omega t}$ convention used throughout this project (Maxwell's equations in this form are standard textbook material – e.g. J. D. Jackson, *Classical Electrodynamics*):

$$\begin{aligned} \text{curl } \mathbf{E} &= -j\omega\mathbf{B} && \text{(Faraday)} \\ \text{curl } \mathbf{H} &= \mathbf{J}_{\text{imp}} + \sigma\mathbf{E} + j\omega\epsilon\mathbf{E} && \text{(Ampere-Maxwell, with} \\ &&& \mathbf{J} = \mathbf{J}_{\text{imp}} + \sigma\mathbf{E} \text{ and} \\ &&& \mathbf{D} = \epsilon\mathbf{E}) \end{aligned}$$

$$\text{div } \mathbf{B} = 0$$

Define the potentials the same way as the rest of this project:

$$\mathbf{B} = \text{curl } \mathbf{A}$$

$\text{div}(\text{curl } \mathbf{A}) = 0$ is an identity, so this automatically satisfies $\text{div } \mathbf{B} = 0$ for *any* choice of $\mathbf{A}$ – gauge freedom enters exactly here and is fixed separately (Phase 03: tree-cotree; Phase 09: Coulomb gauge).

Substituting $\mathbf{B} = \text{curl } \mathbf{A}$ into Faraday's law:

$$\text{curl } \mathbf{E} = -j\omega(\text{curl } \mathbf{A}) \Rightarrow \text{curl}(\mathbf{E} + j\omega\mathbf{A}) = 0$$

A curl-free vector field is the gradient of some scalar, so $\mathbf{E} + j\omega\mathbf{A} = -\text{grad}(\Phi)$ for some scalar potential Φ (the minus sign is the standard electrostatics-potential convention). This gives the second potential definition already used throughout this project:

$$\mathbf{E} = -j\omega\mathbf{A} - \text{grad}(\Phi)$$

1.2 Substituting into Ampere-Maxwell

With $\mathbf{H} = (1/\mu)\mathbf{B} = (1/\mu)\text{curl } \mathbf{A}$:

$$\begin{aligned} \text{curl}[(1/\mu)\text{curl } \mathbf{A}] &= \mathbf{J}_{\text{imp}} + (\sigma + j\omega\epsilon)\mathbf{E} \\ &= \mathbf{J}_{\text{imp}} + (\sigma + j\omega\epsilon)(-j\omega\mathbf{A} - \text{grad}(\Phi)) \end{aligned}$$

$$= J_imp - j*\omega*(\sigma + j*\omega*\epsilon)*A - (\sigma + j*\omega*\epsilon)*\text{grad}(\Phi)$$

Expand the coefficient on A: $-j*\omega*(\sigma + j*\omega*\epsilon) = -j*\omega*\sigma + \omega^2*\epsilon = -(j*\omega*\sigma - \omega^2*\epsilon)$. Moving that term to the left-hand side:

$$\text{curl}[(1/\mu)*\text{curl } A] + (j*\omega*\sigma - \omega^2*\epsilon)*A + (\sigma + j*\omega*\epsilon)*\text{grad}(\Phi) = J_imp \quad (I)$$

This is exactly the first locked-in equation.

1.3 The second equation: current continuity

Taking the divergence of the Ampere-Maxwell law and using $\text{div}(\text{curl } H) = 0$ (an identity, like $\text{div}(\text{curl } A) = 0$ above) gives current continuity in the frequency domain:

$$\text{div}[J_imp + (\sigma + j*\omega*\epsilon)*E] = 0$$

Assumption made explicit here (not stated as a caveat in the roadmap’s one-line equation, so worth recording): this assumes $\text{div}(J_imp) = 0$, i.e. the impressed source current is itself divergence-free (solenoidal) – true for the stranded/massive conductor loop sources this project already commits to in Phase 06 (Dular et al. 2000), and the standard assumption for antenna feed and plane-wave/dipole excitation. A source with genuine free charge accumulation would carry a $-\text{div}(J_imp)$ term on the right-hand side of equation (II) below; that case isn’t needed for either target test problem in Section 7 and is deferred until it is.

Where this could actually matter later (not a near-term risk): every source model Phase 06 plans to build – stranded and massive conductor excitation, per Dular et al. (2000) – is a current-loop or terminal-pair model, divergence-free by construction (a closed circuit has to return the current it sends out; there’s no way to build a physical stranded-coil or massive-conductor source that violates this). Plane-wave/dipole scattering excitation (Phase 07) doesn’t touch this assumption at all, since it’s implemented as an incident field, not a volumetric J_imp . The one place worth flagging on this project’s own roadmap: the **terahertz photoconductive antenna (PCA) gap feed**, already named in docs/THZ_PHASED_ARRAY_SCOPE.md. A PCA gap’s current is generated by ultrafast photoexcitation of carriers, and during that turn-on transient there can be genuine local charge accumulation right at the gap edges – it isn’t obviously a closed-loop current the way a coil winding is. If a PCA gap’s photocurrent is ever modeled as a volumetric J_imp (rather than as a port/boundary excitation), re-check this assumption before trusting the result. The fix, if needed, is a one-term change, not a rework: equation (II) becomes $-\text{div}[(j*\omega*\sigma - \omega^2*\epsilon)*A + (\sigma + j*\omega*\epsilon)*\text{grad}(\Phi)] = -\text{div}(J_imp)$; nothing else in Section 1’s derivation, the DC decomposition in Section 2, or the DOF layout in Section 5 depends on this assumption.

Under that assumption, $\text{div}[(\sigma + j*\omega*\epsilon)*E] = 0$. Substituting $E = -j*\omega*A - \text{grad}(\Phi)$:

$$\begin{aligned} (\sigma + j*\omega*\epsilon)*E &= -j*\omega*(\sigma + j*\omega*\epsilon)*A \\ &\quad - (\sigma + j*\omega*\epsilon)*\text{grad}(\Phi) \\ &= -(j*\omega*\sigma - \omega^2*\epsilon)*A \\ &\quad - (\sigma + j*\omega*\epsilon)*\text{grad}(\Phi) \end{aligned}$$

so current continuity becomes

$$-\text{div}[(j*\omega*\sigma - \omega^2*\epsilon)*A + (\sigma + j*\omega*\epsilon)*\text{grad}(\Phi)] = 0 \quad (II)$$

– the second locked-in equation, with no separate derivation needed beyond the potential substitution already used for (I).

1.4 The locked-in system

$$\begin{array}{l} +-----+ \\ | \text{curl}[(1/\mu) \text{curl } A] + (j*\omega*\sigma - \omega^2*\epsilon) A | \\ | \qquad \qquad \qquad + (\sigma + j*\omega*\epsilon) \text{grad}(\Phi) = J_imp \quad (I) | \\ | | \\ | -\text{div}[(j*\omega*\sigma - \omega^2*\epsilon) A + (\sigma + j*\omega*\epsilon) \text{grad}(\Phi)] = 0 \quad (II) | \end{array}$$

How the two paths share code: (I')-(II') are recovered from (I)-(II) by a single coefficient substitution ($\epsilon \rightarrow 0$, $\Omega_{\Phi} \rightarrow \Omega_c$), not a separate weak form or assembly path. Phase 04's assembly code takes this as a material-property and domain-restriction input, not a branch in the mathematics.

4. Why A-Phi over E-/H-field here (the one-paragraph pitch)

Lee, Lee & Lee (2003), introducing their own A-V (i.e. A-Phi) finite-element formulation, state plainly that it “can be better conditioned than the one obtained by E- or H-field formulation” (Sec. II, citing their own prior work for the underlying result). The mechanism, consistent with the low-frequency breakdown analysis already grounding Phase 05 (Yan 2021, Sec. 1-2): a pure E-field (or H-field) curl-curl formulation has only one family of unknowns, and at low frequency its mass-like term ($\omega^2 \epsilon$, the piece that distinguishes a genuine field from an arbitrary gradient) vanishes quadratically while the curl-curl stiffness term does not – so the *entire* discretized operator inherits a large, ill-conditioned near-null-space (every gradient field) as $\omega \rightarrow 0$, and there is nowhere else in the formulation for that failure to go. Splitting the unknowns into A (solenoidal-plus-gauge content, via curl-curl) and Phi (irrotational content, via a well-conditioned Laplacian-type operator, $\text{div}(\sigma \text{grad}(\Phi))$) moves the low-frequency gradient content out of the ill-conditioned curl-curl block and into Phi's own equation, where it's solved directly rather than fought for as a near-null-space direction inside a single unified unknown. The gauge freedom of A (fixed by tree-cotree or Coulomb gauge, Phases 03/09) is then a smaller, well-understood, well-studied problem rather than a symptom contaminating the whole system. This is the one-paragraph pitch for the whole project: A-Phi doesn't avoid the low-frequency conditioning problem that sinks naive E-/H-field full-wave codes at DC – it relocates that problem to a place (gauge-fixing a vector potential) the literature already knows how to solve well.

5. Degrees-of-freedom layout

5.1 Basis functions (per Phase 02) – UPDATED Sept 2026 (twice): final decision is mixed order

Decision history, kept for the record. Three stances were considered in sequence this month:

1. *Original plan:* matched first order – lowest-order Whitney edge elements for **A** (one unknown per mesh edge, 6 per tet) paired with linear (Lagrange P1) nodal elements for Phi (one unknown per mesh vertex, 4 per tet). Internally consistent (see the discrete identity note below).
2. *Superseded second-order plan:* both fields raised to matched second order – P2 Phi (verified against Jin 2014) together with a *matched* second-order Nedgelec **A**, left as an unverified open item pending Graglia/Wilton/Peterson (1997) and/or Garcia-Castillo et al. (2000).
3. **Final decision (Sept 2026): mixed order.** First-order (lowest-order) Nedgelec **A** together with second-order (P2, 10-node) Phi. This is a deliberate choice, made after the mathematical consequence below was worked out and shown to you explicitly – not an oversight or a default.

Phi: second-order (quadratic, P2) Lagrange nodal elements – 10 nodes per tetrahedron (4 vertices + 6 edge midpoints), **verified directly against J.-M. Jin, *The Finite Element Method in Electromagnetics*, 3rd ed., Wiley, 2014, Ch. 5, Eq. (5.52) and Fig. 5.3** (Sept 2026 – the chapter and figure were read directly, not taken from memory). In tetrahedral volume coordinates $L_1 \dots L_4$ ($\sum L_i = 1$), the ten shape functions are exactly:

$$\begin{aligned} N_i &= (2L_i - 1)L_i && \text{for the 4 vertex nodes, } i = 1..4 \\ N_5 &= 4L_1L_2, & N_6 &= 4L_1L_3, & N_7 &= 4L_1L_4 \\ N_8 &= 4L_2L_3, & N_9 &= 4L_3L_4, & N_{10} &= 4L_2L_4 \end{aligned} \quad (6 \text{ edge-midpoint nodes})$$

This is completely generic finite-element theory – true of any code offering second-order tetrahedra – so it needs no source beyond Jin's textbook and standard FEM theory.

A: the classic first-order (lowest-order) $H(\text{curl})$ -conforming Whitney/Nedgelec tetrahedral edge element – 6 unknowns per tet, one per edge, each representing integral_{edge} $A \cdot d\mathbf{l}$. Basis functions $N_{ij} = L_i \text{grad}(L_j) - L_j \text{grad}(L_i)$, curl constant per tet: $\text{curl}(\mathbf{A}) = \sum_e \mathbf{A}_e \cdot (2 \text{grad}(L_i) \times \text{grad}(L_j))$. This is standard Whitney-element theory, already verified in this project against Bossavit (1998), Ch. 5, Prop. 5.4 (Section 5.1's original citation), and is also **confirmed against your own prior implementation:** BasisFunction.docx (shared Sept 2026), which walks through EdModel.cpp from your university “3dedyaphi” multi-author eddy-current A-Phi

codebase (confirmed not Ansys-derived) – edge-DOF storage (`Edge(e) -> get_ADofVal()`), the `mEdgeSign` orientation convention, and the `CurlFromEdges` routine all match the textbook first-order Nedelec element exactly, with no enrichment or higher-order terms present in the actual discretization.

The “10 points per tet” for $\mathbf{A}$ is a post-processing reconstruction, not a second-order $\mathbf{A}$ basis – worth stating precisely since it’s easy to conflate. The same document shows a separate routine that takes the already-solved 6-edge-DOF field and evaluates/interpolates it onto the same 10-point grid used for Φ (4 vertices + 6 edge midpoints), purely so $\mathbf{A}$ and Φ can be sampled at common points for combined plotting or post-processing: vertex values are reconstructed from a weighted sum of edge DOFs (the exact first-order Whitney formula evaluated at that vertex), mid-edge values are then filled in by a **plain linear average of the two adjacent reconstructed vertex values** (explicitly “the linear part of the 10-node interpolation” in the source document, not a quadratic edge basis), and curl is simply broadcast to all 10 points since it is constant per tet for a first-order element. None of this adds degrees of freedom to $\mathbf{A}$ or changes what goes into the assembled matrices in Section 5.2 below – it is an output/visualization utility, to be implemented later (alongside Phase 08 verification and post-processing) once a solve exists to reconstruct from.

Why the mixed pairing is not “free” – the known consequence, stated and accepted rather than glossed over: the gradient of a P2 vertex shape function, e.g. $\text{grad}[(2\mathbf{Li}-1)\mathbf{Li}] = 4\mathbf{Li}\cdot\text{grad}(\mathbf{Li}) - \text{grad}(\mathbf{Li})$, has a linear (non-constant) part whose Jacobian is $4\cdot\text{grad}(\mathbf{Li}) \otimes \text{grad}(\mathbf{Li})$ – an outer product of a vector with itself, hence a **symmetric** matrix. The first-order Nedelec space’s linear part is exactly the classical $\mathbf{b} \otimes \mathbf{x}$ family (Nedelec 1980’s lowest-order characterization), whose Jacobian is always **skew-symmetric**. A nonzero symmetric matrix can never equal a skew-symmetric one, so $\text{grad}(\text{P2 nodal function})$ is not representable in the first-order edge space in general (this is a from-scratch check, not sourced from any paper – ordinary linear algebra). Concretely: $\text{grad}(\text{P1 subset of first-order edge space})$ holds exactly (the basis for CG = 0 and tree-cotree), but $\text{grad}(\text{P2 subset of first-order edge space})$ does not. Two practical consequences follow, both accepted as part of this decision rather than left as an open question:

- The $\mathbf{A}$ - Φ coupling term’s accuracy is capped by $\mathbf{A}$ ’s lower order – Φ ’s extra edge-midpoint DOFs add representational richness that the coupling into $\mathbf{A}$ cannot fully exploit. This is a known, bounded accuracy trade-off, not a stability or well-posedness problem.
- Tree-cotree gauging (Phase 03) is **unaffected** by this and needs no re-derivation: it operates on $\mathbf{A}$ ’s own edge space and the mesh graph alone (Section 5.3), independent of whatever order Φ uses. Because $\mathbf{A}$ is back to plain first-order, Phase 03’s earlier open item (which only existed because a second-order $\mathbf{A}$ was being considered) no longer applies – the original tree-cotree treatment goes through exactly as first written. Graglia, Wilton & Peterson (1997) and Garcia-Castillo et al. (2000) are therefore **no longer blocking Phase 02** – they remain identified, real, unread candidates for a possible later matched-second-order upgrade to $\mathbf{A}$ (Phase 10 territory), not a dependency for the current mixed-order implementation.

Historical record, from the now-superseded matched-second-order stance (kept because the analysis remains correct and may matter again if $\mathbf{A}$ is later upgraded to matched second order in Phase 10):

Correction to an earlier working assumption (Sept 2026): this decision was originally going to be grounded in Lee, Lee & Lee (2003)’s hierarchical Table I (already cited in Section 4 and docs/REFERENCES.md) as the matched edge/nodal pair. On closer reading, that table is built for a **2-D triangular element** – the paper analyzes 2-D waveguide cross-sections, not 3-D tetrahedra (its own Fig. 1 is a 2-D mesh graph, and its governing equations are the 2-D transverse-field reduction, not the 3-D system this project uses; note it has no “per tet” basis count at all – its Order-1 row gives 3 functions per triangle, one per edge, since a triangle has 3 edges, whereas a generic 3-D lowest-order tetrahedral edge element has 6, one per edge of a tet). The general principle it demonstrates – match the nodal element’s order to the edge element’s order so grad of the nodal space lands inside the edge space – still motivated pulling this decision forward, but the specific formulas have to come from a genuinely 3-D reference instead, hence the two candidates above rather than Lee (2003) itself.

How Lee’s 2-D order-2 pairing actually works (Sept 2026, read directly from the paper, pp. 1898-99): worth recording in detail because it shapes what “matched second order” should mean for the 3-D element once verified. Lee’s own terminology has no “zeroth order” – their text states directly that the classical Whitney edge functions *are* “the first-order vector basis functions” (their term, not an external convention). But the order-2 edge/vector space is **not** simply “order-1 Whitney left as-is, plus a higher-order nodal space bolted on.” The paper states explicitly that the vector basis functions of order 2 and up are split into a gradient part and a rotational-like part, that Table I lists only

the rotational-like part, and that the gradient part “is obtained from the gradients of the second-order and higher order scalar basis functions” in that same table. Concretely, for a triangle:

- order-1 vector (Whitney, kept as a literal subset): 3 functions
- order-2 rotational-like (tabulated directly): 2 functions
- order-2 gradient part = grad of the 3 order-2 scalar functions ($L_i * L_j$): 3 more vector functions, not separately tabulated because they follow mechanically from the scalar column
- total order-2 vector space: $3 + 2 + 3 = 8$ functions, paired with a 6-function order-2 scalar space (3 order-1 L_i + 3 order-2 $L_i * L_j$ – exactly the same dimension as a quadratic/P2 triangle) So order-1 Whitney does survive intact inside the order-2 element, but the full order-2 edge space is larger than order-1 alone specifically *because* it also contains the gradients of the quadratic nodal functions by construction – which is exactly what makes $\text{grad}(\text{order-2 nodal})$ subset of order-2 edge space hold, rather than being incidental. This is a strong hint at the structural shape the verified 3-D element should have, though it is not itself a 3-D formula.

One more data point from the same discussion: Lee’s paper frames the need for tree-cotree (Section V) as arising *only* because the order-1 Whitney part lacks a built-in gradient/rotational split – order 2 and up are already exactly decomposed by construction (per the paragraph above), so Helmholtz splitting is “already fulfilled” there and only the order-1 piece needs it. Read at face value, this suggests tree-cotree’s job in a hierarchical scheme is confined to the order-1 subspace, with higher-order gauge freedom already resolved by the explicit gradient construction. This is a useful precedent but not a transferable result: it comes from a 2-D, single-potential formulation (A_{τ}, ϕ_z , gauged by $A_z = 0$), not this project’s 3-D, two-field $\mathbf{A}$ - Φ system with $\mathbf{A}$ and Φ as independent unknowns coupled through the weak form – so whether the same “tree-cotree only touches the lowest-order piece” structure carries over still depends on how the verified 3-D element (Graglia/Wilton/Peterson 1997 and/or Garcia-Castillo et al. 2000) is actually built.

This historical analysis was written to answer “what would tree-cotree need to gauge if $\mathbf{A}$ were a genuine matched second-order Nedelec element?” – a question that no longer applies to the current (first-order $\mathbf{A}$) decision, per the consequence paragraph above. It is retained here, rather than deleted, in case Phase 10 revisits a matched-second-order $\mathbf{A}$; at that point this is the starting point, not a fresh investigation.

5.2 Raw (ungauged) system size

```

N_A      = N_edges                (full-form: first-order Nedelec has exactly
                                   one DOF per mesh edge, globally -- the
                                   classic Whitney-element count)
      = N_edges(Omega_c union its boundary edges as needed by Phase 03)
                                   (reduced-form, mirroring Phi below)
N_Phi    = N_vertices + N_edges   (full-form: Phi defined everywhere -- P2 has
                                   one DOF per mesh vertex AND one per mesh
                                   edge, since edge-midpoint nodes are shared
                                   by every tet touching that edge)
      = N_vertices(Omega_c) + N_edges(Omega_c)  (reduced-form: Phi confined
                                                    to the conductive region)

N_total_raw = N_A + N_Phi

```

Both formulas are now fully locked in. N_{Φ} follows directly from Jin’s verified 10-node element (Section 5.1) once you account for edge-midpoint nodes being shared between every tetrahedron touching that edge, so each contributes exactly one global DOF, same as a vertex. N_A is the classic first-order Whitney/Nedelec count – exactly one DOF per mesh edge, the same formula the original (pre-second-order-detour) Phase 02 plan always had; it’s back to being simple precisely because Section 5.1’s final decision kept $\mathbf{A}$ at first order.

Every DOF is complex-valued (the frequency-domain system is complex-symmetric per docs/CONDITIONING.md), except at $\omega = 0$ where the imaginary parts are structurally zero (Section 2) – the code keeps the complex type at DC too, so the two regimes share one assembly and solve path rather than a real vs. complex branch.

5.3 Gauge reduction (forward pointer to Phase 03)

The raw system above is singular (Section 2's curl-curl null space argument holds at every frequency, not only $\omega = 0$ – gauge freedom in $\mathbf{A}$ is not a DC-only issue). Phase 03 removes it. Because $\mathbf{A}$ is a plain first-order Nedelec element (Section 5.1's final decision), the classical tree-cotree treatment applies directly and without qualification – there is no longer an open dependency on an unverified second-order element here, unlike the brief period earlier this month when matched-second-order $\mathbf{A}$ was the working plan:

- **Tree-cotree** (default first choice): a spanning tree over the mesh's node/edge graph is selected and its edges removed from the unknown set, leaving $N_{\mathbf{A_gauged}} = N_{\text{edges}} - N_{\text{tree}}$ edge unknowns, where N_{tree} is approximately N_{nodes} minus the number of independently-grounded PEC bodies (Phase 03, step 4).
- **Coulomb gauge** (Phase 09, later): adds a divergence-penalty coupling block instead of removing edges – $N_{\mathbf{A_gauged}} = N_{\mathbf{A}}$ (no edges removed), with the constraint enforced weakly through extra matrix terms rather than DOF elimination.

Phase 01 doesn't choose between these here; it only fixes that gauge reduction happens to the DOF count in Section 5.2, not to the equations in Section 1.

5.4 Boundary and port DOFs (forward pointers, not decided here)

- **PEC boundary** (either target problem's outer conductor, or the sphere's surface in the scattering problem): tangential- $\mathbf{A}$ edges on that boundary are fixed ($\mathbf{n} \times \mathbf{A} = \mathbf{0}$) and removed from the unknown set, per the boundary condition already used in Phase 01's own equations' PEC limit.
- **Phase 07 ABC** (scattering track, outer truncation boundary): boundary edge and node DOFs stay in the unknown set; the ABC enters as an added Robin-type term in the weak form (docs/OPEN_BOUNDARY_ABC.md), not a DOF elimination.
- **Phase 06 port abstraction** (EDA track): not defined yet at Phase 01 – the coaxial/via target problem in Section 7.1 uses a lumped current-source excitation sufficient to exercise Phase 01-05, with the proper port abstraction added in Phase 06.

6. EDA requirements vs. scattering requirements

These two tracks share the equations and DOF layout above but need materially different machinery, per docs/ROADMAP.md Phase 01, step 6:

EDA (Phase 06) needs: - Stranded-conductor and massive-conductor source models (two distinct excitation types, Dular et al. 2000). - Lossy dielectrics and finite-conductivity conductors as material regions. - Thin trace/via geometry handling appropriate to interconnect structures. - A port abstraction (voltage/current or wave-port) for S-parameter/impedance extraction. - The reduced low-frequency variant (Section 3) as a usable, cheaper option for problems where displacement current is negligible.

Scattering (Phase 07) needs: - Plane-wave and dipole incident-field excitation across a real frequency range (not just quasi-static). - An open/radiating outer boundary treatment (the first-order ABC, docs/OPEN_BOUNDARY_ABC.md) – a materially different boundary-condition story from EDA's PEC/port boundaries. - Near-to-far-field / RCS post-processing. - The full all-frequency form (Section 1), since a radiating problem cannot be reduced to the eps-dropped variant without losing the physics being tested.

7. Target test problems

Per docs/ROADMAP.md Phase 01, step 7: one problem per track, chosen so both have closed-form or well-known analytical checks and both will be reused directly by the Phase 08 verification suite.

7.1 EDA target: a coaxial via / power-plane analog

Geometry: a straight cylindrical via of radius $a = 0.1$ mm on-axis inside a cylindrical return/shield conductor of radius $b = 0.5$ mm, length $L = 2$ mm, filled with a lossy dielectric representative of an EDA package ($\epsilon_{\text{ps_r}} = 4.3$,

$\tan(\delta) = 0.02$, i.e. $\sigma = \omega \epsilon'' = \omega \epsilon_r \epsilon_0 \tan(\delta)$ – a frequency-dependent loss tangent model, the standard way to represent lossy FR4-like dielectrics). Excitation: a lumped current source I_0 at one end, the via shorted to the return conductor at the other end (Section 5.4 notes the proper Phase 06 port replaces this lumped source later).

Closed-form checks (standard coaxial-line results, reused directly by Phase 08’s “solenoid/coaxial inductor” benchmark):

$Z_0 = (1 / (2\pi)) * \sqrt{\mu / \epsilon} * \ln(b/a)$	(characteristic impedance)
$L' = (\mu / (2\pi)) * \ln(b/a)$	(inductance per unit length)
$C' = (2\pi\epsilon) / \ln(b/a)$	(capacitance per unit length)
$R_{DC} = 1 / (\sigma_{cond} * \pi * a^2)$	(DC resistance per unit length, conductor conductivity σ_{cond})

Why this problem exercises both paths: at DC and low frequency (up to roughly where the dielectric’s displacement current becomes comparable to its conduction current, or where the via’s electrical length is small compared to the exciting wavelength), the reduced low-frequency variant (Section 3) should reproduce R_{DC} and L' directly. At higher frequency, where displacement current in the lossy dielectric is no longer negligible, only the full all-frequency form (Section 1) stays valid – exactly the regime split this whole project is built around, tested on one geometry instead of two.

7.2 Scattering target: a PEC sphere at a real operating frequency (Mie benchmark)

Geometry: a perfectly-conducting sphere of radius $a = 1$ cm in free space, illuminated by a normally-incident plane wave. **Operating frequency:** chosen so $k*a = 1$ (the resonance/Mie regime, not the Rayleigh low-frequency limit) – $f = c / (2\pi*a) \approx 4.77$ GHz. This is deliberately a “real operating frequency” per docs/ROADMAP.md Phase 01, step 7’s own wording, distinct from the separate $k*a \rightarrow 0$ low-frequency Mie-limit regression case Phase 08 already commits to running – Phase 01 locks in the general full-wave problem statement here; Phase 08 formalizes both regimes (resonance and the low-frequency limit) as automated regression tests on the same analytical family.

Closed-form check: the classical Mie series for a PEC sphere gives the exact scattered field, backscatter/bistatic radar cross section, and induced surface current – standard closed-form electromagnetics, the same ground-truth family Phase 08 already names (“low-frequency sphere scattering (Mie limit)”).

What this exercises that Section 7.1 doesn’t: the Phase 07 open-boundary ABC (docs/OPEN_BOUNDARY_ABC.md) and the full radiating all-frequency system at a genuinely non-quasi-static frequency – the coaxial/via problem never needs an open boundary at all.

7.3 Note on the terahertz phased-array case (not a third target yet)

Per docs/ROADMAP.md Phase 01, step 8: a THz phased-array feed (a photoconductive gap, HEMT gate, or via transition at tens-to-hundreds of nm) driving an electrically large radiating aperture pushes the same spatial-scale argument behind Section 4 past a $10^4:1$ ratio – sharper than Section 7.1’s via, but the identical underlying mechanism. This is not a third target problem for Phase 01-05; Phases 06/07 add the array-specific machinery (multi-port active impedance, Floquet boundary conditions) that a third target problem would need first. See docs/THZ_PHASED_ARRAY_SCOPE.md.

Status against Phase 01’s “Ready for 02” checklist

Per docs/ROADMAP.md: “the full (all-frequency) equations, the reduced low-frequency variant kept as an option, the DOF layout, and the two target test problems ... are written down in docs/.” All four are now in this document: Section 1 (full equations, derived from Maxwell’s equations rather than only asserted), Section 3 (reduced variant), Section 5 (DOF layout), and Section 7 (both target test problems).

Sept 2026 update (final): Section 5.1’s basis-function choice went through two revisions this month and settled on a **mixed-order pairing:** first-order (lowest-order) Whitney/Nedelec **A** together with second-order (P2, 10-node) **Phi**. Both sides are now fully verified and ready for Phase 02 – **Phi** against Jin (2014), **A** against Bossavit (1998) and

your own confirmed prior implementation (`BasisFunction.docx / EdModel.cpp`, from your university “3dedyaphi” codebase). The known consequence of the mixed pairing – `grad(P2 Phi)` not being exactly representable in the first-order **A** edge space, capping A-Phi coupling accuracy at first order – is documented and accepted, not left open. Phase 03’s tree-cotree writeup is **not** blocked: because **A** stayed first-order, the classical treatment applies unchanged (Section 5.3). Graglia, Wilton & Peterson (1997) and Garcia-Castillo et al. (2000) remain identified as candidates for a possible later matched-second-order upgrade to **A** (Phase 10), but are no longer a Phase 02 dependency.

References

- J. D. Jackson, *Classical Electrodynamics*, Wiley – standard textbook source for Maxwell’s equations and the A/Phi potential definitions used in Section 1 (general background, not specific to any one implementation).
- Y. Zhao, W. N. Fu, “A New Stable Full-Wave Maxwell Solver for All Frequencies,” *IEEE Trans. Magn.*, vol. 53, no. 6, 2017 – basis for keeping the eps terms (Section 1.4); already cited in `docs/REFERENCES.md`.
- S. Yan, “Continuous Discontinuous Galerkin Method for Electromagnetic Simulations Based on an All Frequency Stable Formulation,” *Progress In Electromagnetics Research M*, vol. 106, 2021 – same role as above, and the low-frequency-breakdown mechanism referenced in Section 4.
- S.-C. Lee, J.-F. Lee, R. Lee, “Hierarchical Vector Finite Elements for Analyzing Waveguiding Structures,” *IEEE Trans. Microwave Theory Tech.*, vol. 51, no. 8, 2003 – Sec. II is the direct source for the A-V-vs-E/H conditioning claim in Section 4 (verified against the actual paper text, Sept 2026, not just the summary already in `docs/REFERENCES.md`). **Scope correction (Sept 2026):** this paper’s Table I hierarchical basis set (used in an earlier working draft of Section 5.1 as the model for matching a second-order **A** to a second-order Phi) was re-checked against the paper text and found to be a 2-D triangular-element scheme for waveguide cross-sections (Fig. 1), not a 3-D tetrahedral scheme – it no longer sources the A-side basis-function decision below; it is retained here only for the A-V-vs-E/H conditioning claim, which is unaffected.
- J.-M. Jin, *The Finite Element Method in Electromagnetics*, 3rd ed., Wiley, 2014, Chapter 5 “Three-Dimensional Finite Element Analysis” – direct, verified source for the second-order (10-node) nodal tetrahedral shape functions used for Phi in Section 5.1 (Eq. 5.52, Fig. 5.3; chapter and figure read directly, pages checked against the shared PDF, Sept 2026). This chapter covers only nodal/Lagrange elements – it contains no edge, Whitney, vector-basis, H(curl), or Nedgelec material, so it cannot source the A-side element below.
- R. D. Graglia, D. R. Wilton, A. F. Peterson, “Higher Order Interpolatory Vector Bases for Computational Electromagnetics,” *IEEE Trans. Antennas Propag.*, vol. 45, no. 3, 1997 – candidate primary source for the matched second-order tetrahedral edge element for **A** (Section 5.1). Identified via search (confirmed to exist, IEEE Xplore document 558649) but **not yet read**; do not implement Phase 02’s A-side basis functions from this title alone.
- L. E. Garcia-Castillo et al., “Second-order Nedgelec tetrahedral element for computational electromagnetics,” *Int. J. Numer. Model.*, vol. 13, no. 2-3, 2000 – second candidate primary source for the same A-side element, title is an exact match for what is needed. Identified via search (confirmed to exist, Wiley Online Library) but **not yet read**; no longer a Phase 02 dependency now that **A** stayed first-order (Sept 2026) – retained as a candidate for a possible Phase 10 matched-second-order upgrade.
- Nastaran Hendijani, `BasisFunction.docx` (shared Sept 2026), summarizing `EdModel.cpp` from her university “3dedyaphi” multi-author eddy-current A-Phi research codebase (confirmed not Ansys-derived) – direct source for the final first-order Nedgelec **A** decision in Section 5.1: edge-DOF storage, the `mEdgeSign` orientation convention, and the `Cur1FromEdges` routine, all matching the textbook first-order Whitney/Nedgelec element exactly. The same document’s separate 10-point reconstruction routine (vertex values from edge DOFs, mid-edge values by linear averaging) is a post-processing/visualization utility, not part of the Section 5.2 discretization.
- P. Dular et al., “Dual magnetodynamic formulations and their source fields associated with massive and stranded inductors,” *IEEE Trans. Magn.*, vol. 36, no. 4, 2000 – basis for the reduced low-frequency variant (Section 3) and the EDA source models it shares with Phase 06; already cited in `docs/REFERENCES.md`.